1 Matrix of an isomorphism

Let V and W be finite-dimensional vector spaces and suppose $T : V \to W$ is an isomorphism. Then $\dim V = \dim W = n$, so $[T]$ and $[T^{-1}]$ are $n \times n$ matrices. Since $T^{-1}T = \mathrm{id}_V$ and $TT^{-1} = \mathrm{id}_W$ and $[\mathrm{id}_V] = [\mathrm{id}_W] = I$ (the $n \times n$ identity matrix), we have

$$[T^{-1}T] = [T^{-1}][T] = I, \text{ and}$$
$$[TT^{-1}] = [T][T^{-1}] = I.$$

Definition 1.1 (Invertible matrix)**.** Let A is an $n \times n$ matrix. If there exists an $n \times n$ matrix B such that $AB = BA = I$, then A is said to be *invertible* or *non-singular* and B is said to be the *inverse* of A, which we denote by $B = A^{-1}$. A matrix which is not invertible is called *non-invertible* or *singular*.

The matrix of an isomorphism of finite-dimensional vector spaces is therefore an invertible matrix.

Exercise 1.2. Prove that the inverse of a matrix A, if it exists, is unique.

Exercise 1.3. Prove that if A and B are invertible $n \times n$ matrices, then AB is an invertible $n \times n$ matrix with $(AB)^{-1} = B^{-1}A^{-1}$. Similarly, any finite product $A_1 \cdot A_k$ of invertible $n \times n$ matrices is invertible with $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$.

2 Elementary Matrices

Definition 2.1 (Elementary matrix)**.** An $n \times n$ matrix is said to be an *elementary matrix* if it can be obtained from the $n \times n$ identity matrix by a single elementary row operation.

Example 2.2. Let c be a constant. Verify that the complete list of 2×2 elementary matrices is

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}, \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} (c \neq 0), \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} (c \neq 0).$$

Proposition 2.3. *If $e : M^{n\times n}(F) \to M^{n\times n}(F)$ is an elementary row operation and E the $n \times n$ elementary matrix $E = e(I)$, then for every $m \times n$ matrix A, $e(A) = EA$.*

Proof. We consider each type of elementary row operation separately.

(i) Suppose $r \neq s$ and let e be the elementary row operation which replaces row r by row r plus c times row s. We need to show that $EA = e(A)$. We can write the elements of the corresponding elementary matrix E as [1]

$$E_{ik} = \begin{cases} \delta_{ik}, & i \neq r \\ \delta_{rk} + c\delta_{sk}, & i = r. \end{cases}$$

Then, if $i \neq r$,

$$(EA)_{ij} = \sum_{k=1}^{m} E_{ik}A_{kj} = \sum_{k=1}^{m} \delta_{ik}A_{kj} = A_{ij}$$

and for $i = r$,

$$(EA)_{rj} = \sum_{k=1}^{m} E_{rk}A_{kj} = \sum_{k=1}^{m} (\delta_{rk} + c\delta_{sk})A_{kj} = A_{rj} + cA_{sj}$$

which shows $EA = e(A)$, as desired.

(ii) Suppose $r \neq s$ and let e be the elementary row operation which exchanges rows r and s. The elements of the corresponding elementary matrix are given by

$$E_{ik} = \begin{cases} \delta_{ik}, & i \neq r, s \\ \delta_{rk}, & i = s \\ \delta_{sk}, & i = r. \end{cases}$$

Then, if $i \neq r, s$,

$$(EA)_{ij} = \sum_{k=1}^{m} E_{ik}A_{kj} = \sum_{k=1}^{m} \delta_{ik}A_{kj} = A_{ij}$$

if $i = r$,

$$(EA)_{rj} = \sum_{k=1}^{m} E_{rk}A_{kj} = \sum_{k=1}^{m} \delta_{sk}A_{kj} = A_{sj}$$

[1] In the following δ_{ij} denotes the ij-component of the identity matrix; that is, $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

and if $i = s$,

$$(EA)_{sj} = \sum_{k=1}^{m} E_{sk}A_{kj} = \sum_{k=1}^{m} \delta_{rk}A_{kj} = A_{rj}$$

which shows $EA = e(A)$.

(iii) Let e be the elementary row operation which multiplies rows r by c, $c \neq 0$. We need to show that $EA = e(A)$. The corresponding elementary matrix can be written as

$$E_{ik} = \begin{cases} \delta_{ik}, & i \neq r \\ c\delta_{rk}, & i = r. \end{cases}$$

Then, if $i \neq r$,

$$(EA)_{ij} = \sum_{k=1}^{m} E_{ik}A_{kj} = \sum_{k=1}^{m} \delta_{ik}A_{kj} = A_{ij}$$

if $i = r$,

$$(EA)_{rj} = \sum_{k=1}^{m} E_{rk}A_{kj} = \sum_{k=1}^{m} (c\delta_{rk})A_{kj} = cA_{rj}$$

which shows $EA = e(A)$. □

Exercise 2.4. (a) Let $E = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$ be the elementary matrix obtained from the 2×2 identity matrix by replacing $R2 \mapsto R2 - 3R1$. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Show that the matrix product EA is equal to the result of performing the row replacement $R2 \mapsto R2 - 3R1$ on A.

(b) Let $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ be the elementary matrix obtained from the 2×2 identity matrix by the interchange $R1 \leftrightarrow R2$. Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. Show that the matrix product EA is equal to the result of interchanging the two rows of A.

Corollary 2.5. *Let A and B be $m \times n$ matrices. Then B is row-equivalent to A if and only if $B = PA$, where P is a product of $m \times m$ elementary matrices.*

Proof. ($\Longrightarrow$) Suppose B is row-equivalent to A. Let $E_1, E_2, \cdots, E_s$ be the elementary matrices corresponding to some sequence of elementary row operations which carries A into B. Then $B = (E_s \cdots E_2 E_1)A$.

($\Longleftarrow$) Suppose $B = PA$ where $P = E_s \cdots E_2 E_1$ and the E_i are $m \times m$ elementary matrices. Then $E_1 A$ is row-equivalent to A, and $E_2(E_1 A)$ is row-equivalent to $E_1 A$, so by transitivity $E_2 E_1 A$ is row-equivalent to A. It follows by induction that $(E_s \cdots E_1)A$ is row-equivalent to A. □

Proposition 2.6. *An elementary matrix E is invertible. The inverse of E is the elementary matrix of the same type that transforms E back into I.*

Proof. Let E be the elementary matrix corresponding to the elementary row operation e. If e' is the inverse operation of e and $E' = e'(I)$, then $EE' = e(E') = e(e'(I)) = I$, so E is invertible and $E' = E^{-1}$. □

Example 2.7. (a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -c \\ 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -c & 1 \end{bmatrix}$

(d) When $c \neq 0$, $\begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{1}{c} & 0 \\ 0 & 1 \end{bmatrix}$, $\quad \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{c} \end{bmatrix}$.

3 Invertible Matrix Theorem

Theorem 3.1 (Invertible matrix theorem)**.** *If A is an $n \times n$ matrix, the following are equivalent.*

(i) A is invertible.

(ii) A is row-equivalent to the $n \times n$ identity matrix.

(iii) A is a product of elementary matrices.

Proof. We will show that $(i) \Longrightarrow (ii) \Longrightarrow (iii) \Longrightarrow (i)$. Assume (i) is true. Let R be the RREF of A. Then $R = E_k \cdots E_2 E_1 A$, where $E_1, \cdots, E_k$ are elementary matrices. Since each E_j is invertible, we have $A = E_1^{-1} \cdots E_k^{-1} R$. Since products of invertible matrices are invertible, we see that A is invertible if and only if R is invertible. Since R is a square matrix in RREF, R is invertible iff it is the $n \times n$ identity matrix (otherwise, it would have a row of zeros and any such matrix is singular). Hence, $(i) \Longrightarrow (ii)$. (iii) follows immediately from (ii). (i) follows immediately from (iii) by Exercise 1.3. □

Corollary 3.2 (How to compute A^{-1})**.** *If A is an invertible $n \times n$ matrix and if a sequence of elementary row operations reduces A to the identity, then that same sequence of operations when applied to I yields A^{-1}.*

Proof. If $E_k \cdots E_1 A = I$, where each E_j is an elementary matrix, then multiplying both sides on the right by A^{-1} gives $E_k \cdots E_1 I = A^{-1}$. □

Corollary 3.2 gives an algorithm to compute A^{-1}: we simply row reduce the augmented matrix $[A|I]$. If A is row-equivalent to I, then $[A|I]$ is row-equivalent tp $[I|A^{-1}]$. Otherwise, A does not have an inverse.

Example 3.3. We find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$ by performing Gauss-Jordan elimination on the augmented matrix

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right],$$

which gives

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & 4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right]$$

so $A^{-1} = \begin{bmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & 4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{bmatrix}$.

Exercise 3.4. Find the inverse of the matrix $A = \begin{bmatrix} 3 & 1 & -2 \\ 1 & 0 & 2 \\ -1 & -2 & 4 \end{bmatrix}$, if it exists.

Exercise 3.5. Prove that a linear system $Ax = y$ of n equations in n unknowns has a unique solution if and only if A is invertible.

Exercise 3.6. Consider the linear system

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5x_2 + 3x_3 &= 3 \\ x_1 + 8x_3 &= 17. \end{aligned}$$

The coefficient matrix is $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$, which has inverse $A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$. Use A^{-1} to find the unique solution of the system.